CT-KAE on transonic flow: reference and back001

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Models

Both are the same continuous-time Koopman autoencoder. A CNN encoder maps two input frames ($\Delta t = 0.1$) and the Mach number to $z \in \mathbb{R}^{256}$. The latent evolves exactly linearly, $z(\tau) = e^{\tau A(\mathrm{Ma})} z_0$, with A block-diagonal: 128 damped oscillators, decay $\gamma \geq 0$. A FiLM-conditioned CNN decoder maps z back to the four fields. 24.7 M parameters; trained 20k steps on the 33 training simulations to reconstruct and to predict 8 frames ahead, with latent consistency. **back001** adds backward latent consistency at weight 0.01. Three seeds each; every number below is a mean over seeds or a 3-seed average, as stated.

Main results

Table 1: Forecasting (MSE $\times 10^{-3}$, all frames; LSiM $\times 10^{-1}$), data assimilation (relative L_2 of the analysis at t_0; df1 headline: 12 problems, 2000 iterations; sweeps: mean over the points of six sweeps that every method was run on, 31 at $\delta_f = 1$ and 37 at $\delta_f = 9$), and inference cost (Tra_long, 240 steps, batch 1, RTX 4090). Lower is better throughout. Post-processing (MSE-optimal averaging) and the sharpened mirror average (non-blurring) are described in the next two sections; the effect of post-processing on DA is in Table 5. Original KAE: final checkpoint; the published DA numbers used a checkpoint selected on the DA test set.

	MSE			LSiM			DA error			ms per
	ext	int	long	ext	int	long	df1 head.	df1 sweeps	df9 sweeps	step
ACDM	2.14	2.92	22.63	1.29	1.38	3.83	0.006	0.049	0.258	142.47
U-Net$_{ut}$	1.60	1.48	22.18	1.09	0.95	3.81	0.215	0.208	0.270	6.66
U-Net$_{tn}$	1.37	1.77	22.38	1.05	1.05	3.92	–	–	–	7.06
Original KAE (final ckpt)	2.05	11.05	18.47	–	–	–	0.029	0.029	0.103	0.80
Reference	1.91	3.37	16.08	1.41	1.71	4.06	0.029	0.046	0.095	0.27
Reference + post-proc.	1.31	2.45	11.87	1.73	1.86	7.24	–	–	–	7.68
Reference + sharp. mirror	1.34	2.57	15.81	1.20	1.53	4.07	–	–	–	–
back001	2.03	3.04	15.11	1.49	1.71	4.88	0.024	0.031	0.088	0.27
back001 + post-proc.	1.46	2.46	11.97	1.90	1.81	7.47	–	–	–	7.68
back001 + sharp. mirror	1.53	2.59	14.88	1.33	1.57	4.91	–	–	–	–

- **Long rollouts:** at 240 steps both CT-KAEs, without post-processing, have 27–33% lower MSE than ACDM and the U-Nets and 13–18% lower than the original KAE (46–47% and 35–36% with it), and stay correlated with the truth over 1000 steps (Fig. 3).
- **Data assimilation:** both are at least 2× more accurate than the U-Net and FNO at every lead. ACDM + SDA wins at $\delta_f \leq 2$. Against the original KAE, back001 is equal at short leads and better from $\delta_f = 8$; the reference is worse at short leads (Fig. 1).
- **Short horizon:** without post-processing the U-Nets are best on ext and int. With it the reference is best on ext; the U-Nets remain best on int. The sharpened mirror average gives almost the same MSE without blurring: on ext the reference has lower MSE than every baseline (1.34) and better LSiM than ACDM (1.20 against 1.29); the U-Nets keep the best LSiM.
- **back001 vs reference:** back001 is better on DA, int and long, worse on ext. Post-processing removes most of the forecasting differences; the DA advantage remains.

- **Sharpness:** even without post-processing the CT-KAE is smoother than the baselines (worse LSiM, spectra too shallow). Post-processing improves MSE by blurring further (Table 3). The sharpened mirror average does not blur: LSiM improves on ext and int, and on long it stays at the single decode's level (Table 4).

Post-processing

All steps act at inference on a trained model. Settings were fixed before the test sets were scored: mixture width and shrinkage on training rollouts, mirror averaging and the choice of steps on two proxy models trained with Mach 0.53–0.54 and 0.73–0.77 held out.

- **Mixture readout.** Average of 5 decodes of the same latent flow run at time scales $1 + \sigma\xi_q$ (Gauss–Hermite, $\sigma \approx 0.04$). Hedges the phase drift of the wake.
- **Mirror averaging.** The flow is (nearly) symmetric about the centre line: flip the input vertically with $v_y \to -v_y$, forecast, flip back, average with the plain forecast.
- **Climatology shrinkage.** Per lead and field, pull the forecast towards the Mach-interpolated training mean by a factor fitted on training rollouts. Affects long leads only.
- **Seed averaging.** Average the forecasts of three independently trained seeds.
- **Cost.** The full pipeline is 30 decodes per step: 7.7 ms per step against 0.27 for a single decode, about the cost of a U-Net and 18× cheaper than ACDM. Peak memory 0.9 GB.
- **Cost in sharpness.** Every step is an average, which MSE rewards and sharpness does not: LSiM and the spectral band errors get worse at every scale (Table 3). Use it when the target is MSE, not realistic fields.
- **Tested and rejected** on the proxies: test-time refinement of z_0 on the input frames (always worse), a mixture over neighbouring Mach numbers (no effect), lead-dependent mixture width, Wiener filtering by wavenumber (no better than one factor per lead and field).

Table 2: Each post-processing step added in turn: MSE $\times 10^{-3}$ (rows 1–4: mean over the 3 seeds; row 5: one 3-seed average), decoder calls per predicted step and wall clock per step.

		ext	int	long	decodes/step	ms/step
Reference	single decode	1.91	3.37	16.08	1	0.27
	+ mixture	1.77	3.16	13.03	5	1.28
	+ mirror	1.40	2.58	12.74	10	2.56
	+ shrinkage	1.40	2.58	11.98	10	2.56
	+ 3-seed average	1.31	2.45	11.87	30	7.68
back001	single decode	2.03	3.04	15.11	1	0.27
	+ mixture	1.88	2.89	13.45	5	1.28
	+ mirror	1.56	2.58	13.06	10	2.56
	+ shrinkage	1.56	2.57	12.07	10	2.56
	+ 3-seed average	1.46	2.46	11.97	30	7.68

Table 3: What post-processing costs in sharpness. MSE $\times 10^{-3}$ on Tra_long; LSiM $\times 10^{-1}$; relative error of the $|v|$ spectrum on Tra_long (240 steps) in the low ($k \leq 4$), mid (5–12) and high (13–32) bands, and its mid-band slope. Lower is better except the slope, which should match the ground truth.

	MSE	LSiM			spectrum, Tra_long			
	long	ext	int	long	low	mid	high	slope
ACDM	22.63	1.29	1.38	3.83	0.06	0.11	0.04	-3.27
U-Net$_{ut}$	22.18	1.09	0.95	3.81	0.05	0.08	0.03	-3.38
Reference	16.08	1.41	1.71	4.06	0.31	0.34	0.20	-2.53
Reference + post-proc.	11.87	1.73	1.86	7.24	0.42	0.57	0.47	-2.36
Reference + sharp. mirror	15.81	1.20	1.53	4.07	0.30	0.37	0.24	-2.46
back001	15.11	1.49	1.71	4.88	0.28	0.41	0.24	-2.41
back001 + post-proc.	11.97	1.90	1.81	7.47	0.41	0.58	0.46	-2.37
back001 + sharp. mirror	14.88	1.33	1.57	4.91	0.27	0.38	0.26	-2.42
Ground truth								-3.30

Non-blurring post-processing: sharpened mirror average

The aim is to lower the error without blurring.

- **Readout.** Per seed, average the single decode and the mirrored single decode (no mixture). Then rescale the anomaly of the average about the Mach-interpolated training mean, per frame, field and radial wavenumber, to the mean band power of the two single decodes. The phases of the average are kept and the gain is clipped to $[1, 5]$; on the proxies the clip never binds within 58 leads. With 3 seeds, all 6 single decodes are averaged and their mean power is the target. Cost: 6 decodes per step and one FFT pair per frame.
- **Selection.** Candidates were scored on the proxies' held-out Mach numbers (7 simulations; 28 windows of 58 leads and 7 of 238): the mirror, mixture and mixture + mirror averages; the latent average of the mixture nodes; re-sharpening of the mirror and mixture + mirror averages (gain exponent 0.5 or 1, clip 2 or 5); Fourier amplitude averaging with the phase of the mean; and rank-preserving quantile mapping. The rule was fixed before scoring: LSiM no worse than the single decode and band power at most 0.05 dex lower, on both proxies and both horizons; then lowest MSE. Strictly, no candidate passes, because at 238 leads every average raises LSiM. The sharpened mirror average is the only one within 1.4 SE everywhere (reference proxy, long: LSiM $+0.07 \pm 0.05$). At 58 leads it lowers MSE by 9% and LSiM by 0.10–0.11 on both proxies. The proxies have one seed, so seed averaging is not validated there.
- **Test** (Table 4; per seed, paired against the single decode, SE over 3 seeds). On ext and int, MSE falls by 10–23% per seed (15–30% with the 3-seed average) and LSiM by 0.10–0.16; the high-band power ratio stays within 0.03 dex of the single decode. On long, MSE falls by 0.19–0.26 (1–2 SE) and LSiM rises by 0.05 (within 1 SE). Over 1000 steps the mean v_x correlation is 0.705 (reference) and 0.699 (back001), against 0.691 and 0.682 for the single decode and 0.730 and 0.722 for post-processing (Fig. 3).
- **Against post-processing** (3 seeds, reference): MSE 1.34 / 2.57 / 15.81 against 1.31 / 2.45 / 11.87 (ext / int / long); LSiM 1.20 / 1.53 / 4.07 against 1.73 / 1.86 / 7.24. Use the sharpened mirror average when realistic fields matter. The long-horizon MSE gain of post-processing comes only from blurring.
- **Rejected.** The latent average of the mixture nodes equals the single decode at 58 leads and blurs at 238. Sharpening the mixture + mirror average keeps the mixture's loss at long leads. Fourier amplitude averaging is close to sharpening but slightly worse. Quantile mapping fixes amplitudes, not scales, so the high band still loses power. In an earlier test-set screen, x-shift-aligned averaging and the medoid member gained nothing. Refinement with the trained ACDM (SDEdit) was not run: each frame would need part of ACDM's sampler (142 ms per step in full), up to $\sim$500$\times$ a single decode.

Table 4: Sharpened mirror average, each step added in turn. MSE $\times 10^{-3}$, LSiM $\times 10^{-1}$, and the mean $\log_{10}$ power ratio (prediction / truth) of the $|v|$ spectrum in the high band (13–32; 0 is exact, negative is smoother). Rows 1–3: mean over 3 seeds; row 4: one 3-seed average; row 5: paired change of row 3 against row 1, $\pm$ SE over seeds.

		MSE			LSiM			high-band log ratio			decodes	ms
		ext	int	long	ext	int	long	ext	int	long	/step	/step
Reference	single decode	1.91	3.37	16.08	1.41	1.71	4.06	-0.01	-0.05	-0.13	1	0.27
	+ mirror	1.44	2.66	15.20	1.33	1.65	4.34	-0.04	-0.11	-0.19	–	–
	+ sharpen	1.48	2.74	15.82	1.25	1.57	4.11	-0.02	-0.08	-0.16	–	–
	+ 3-seed average	1.34	2.57	15.81	1.20	1.53	4.07	-0.02	-0.08	-0.16	–	–
	sharpen vs single	-0.43 ± 0.11	-0.64 ± 0.05	-0.26 ± 0.25	-0.16 ± 0.00	-0.14 ± 0.01	$+0.05 \pm 0.04$	-0.01 ± 0.00	-0.03 ± 0.00	-0.02 ± 0.00		
back001	single decode	2.03	3.04	15.11	1.49	1.71	4.88	-0.01	-0.05	-0.14	1	0.27
	+ mirror	1.61	2.67	14.24	1.47	1.64	5.26	-0.05	-0.10	-0.19	–	–
	+ sharpen	1.66	2.72	14.92	1.37	1.61	4.93	-0.03	-0.07	-0.16	–	–
	+ 3-seed average	1.53	2.59	14.88	1.33	1.57	4.91	-0.02	-0.07	-0.17	–	–
	sharpen vs single	-0.37 ± 0.05	-0.32 ± 0.09	-0.19 ± 0.10	-0.12 ± 0.02	-0.10 ± 0.02	$+0.05 \pm 0.05$	-0.02 ± 0.00	-0.02 ± 0.00	-0.02 ± 0.01		

In data assimilation only mirror averaging applies: 4D-Var is run once in the plain frame and once in the mirrored frame, and the two analyses are averaged. It cuts the analysis error by 10–16% for both models at twice the cost (Table 5).

The paper's transonic experiments, re-run

Short horizon. The U-Nets are best up to $t = 30$. On ext the post-processed reference is best from $t = 45$. On int the U-Net$_{ut}$ stays best; at $t = 60$ the post-processed CT-KAEs are second (5.6–5.7 against 4.9), ahead of ACDM (8.5) and the original KAE (15.8).

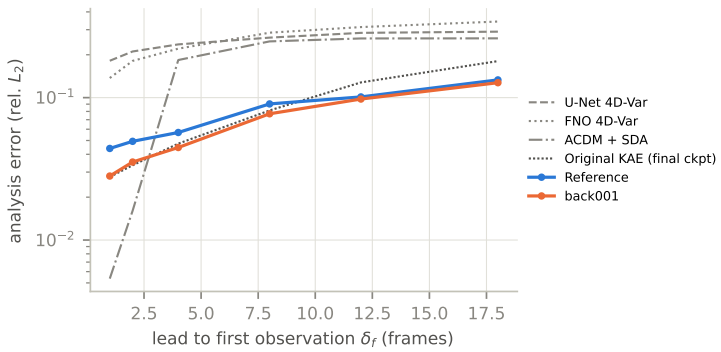


Figure 1: Analysis error against the lead δ_f to the first observation, transonic flow, $\delta_l = 25$, 5 observations, 500 iterations, 8 problems per point.

Table 5: Mirror averaging in 4D-Var: analysis error, seed 2 of each model, same problems as Table 1.

		df1 head.	df1 sweeps	df9 sweeps
Reference	plain	0.030	0.047	0.098
	mirror	0.026	0.042	0.088
	change	-13%	-10%	-10%
back001	plain	0.024	0.032	0.092
	mirror	0.021	0.027	0.082
	change	-16%	-15%	-11%

Table 6: MSE ($\times 10^{-3}$, all fields) at $t = 15, 30, 45, 60$ of the 60-step rollouts. CT-KAE rows without post-processing: mean over 3 seeds. Best per column in bold.

	Tra_ext				Tra_int			
t	15	30	45	60	15	30	45	60
ACDM	0.40	1.36	3.51	6.51	0.43	2.07	5.13	8.46
U-Net$_{ut}$	0.14	0.92	2.70	5.14	0.17	0.96	**2.60**	**4.90**
U-Net$_{tn}$	**0.12**	**0.74**	2.44	4.50	**0.12**	**0.82**	3.00	6.87
Original KAE	0.44	1.10	2.42	3.92	1.27	5.30	9.75	15.78
Reference	0.36	1.54	3.17	4.84	0.64	2.90	5.58	8.48
Reference + post-proc.	0.25	1.05	**2.18**	**3.32**	0.49	2.20	4.11	5.64
Reference + sharp. mirror	0.25	1.08	2.24	3.34	0.49	2.22	4.29	6.02
back001	0.36	1.64	3.36	5.30	0.60	2.64	5.10	7.48
back001 + post-proc.	0.25	1.14	2.44	3.79	0.47	2.20	4.21	5.73
back001 + sharp. mirror	0.25	1.14	2.54	4.00	0.47	2.23	4.45	6.22

Table 7: The latent ODE $\dot{y} = Ay$ integrated numerically instead of $e^{\tau A}$ (seed 0, single decode). MSE $\times 10^{-3}$.

	integrator	ext	int	long
Reference	$e^{\tau A}$ (exact)	1.61	3.32	16.66
Reference	RK4, $\Delta t = 0.1$	1.61	3.32	16.66
Reference	RK4, $\Delta t = 0.3$	1.63	3.28	16.34
Reference	RK4, $\Delta t = 0.5$	1.92	3.22	14.58
Reference	midpoint, $\Delta t = 0.1$	1.77	3.49	17.10
Reference	Euler, $\Delta t = 0.05$	2.28	3.76	24.82
Reference	Euler, $\Delta t = 0.1$	6.94	6.11	diverged
Reference	Dopri5 (adaptive)	1.61	3.32	16.66
back001	$e^{\tau A}$ (exact)	1.98	2.91	14.57
back001	RK4, $\Delta t = 0.1$	1.98	2.91	14.57
back001	RK4, $\Delta t = 0.3$	2.00	2.90	14.46
back001	RK4, $\Delta t = 0.5$	2.29	2.97	14.17
back001	midpoint, $\Delta t = 0.1$	2.02	3.01	14.71
back001	Euler, $\Delta t = 0.05$	2.61	3.71	diverged
back001	Euler, $\Delta t = 0.1$	8.81	12.37	diverged
back001	Dopri5 (adaptive)	1.98	2.91	14.57

Table 8: Unseen time step: MSE at leads on the training grid (0.1 s) and at the half-steps between them (0.05 s, never trained on). Mean over 3 seeds.

	ext		int	
	0.1	0.05	0.1	0.05
Reference	1.98	1.99	3.49	3.51
back001	2.10	2.11	3.14	3.16

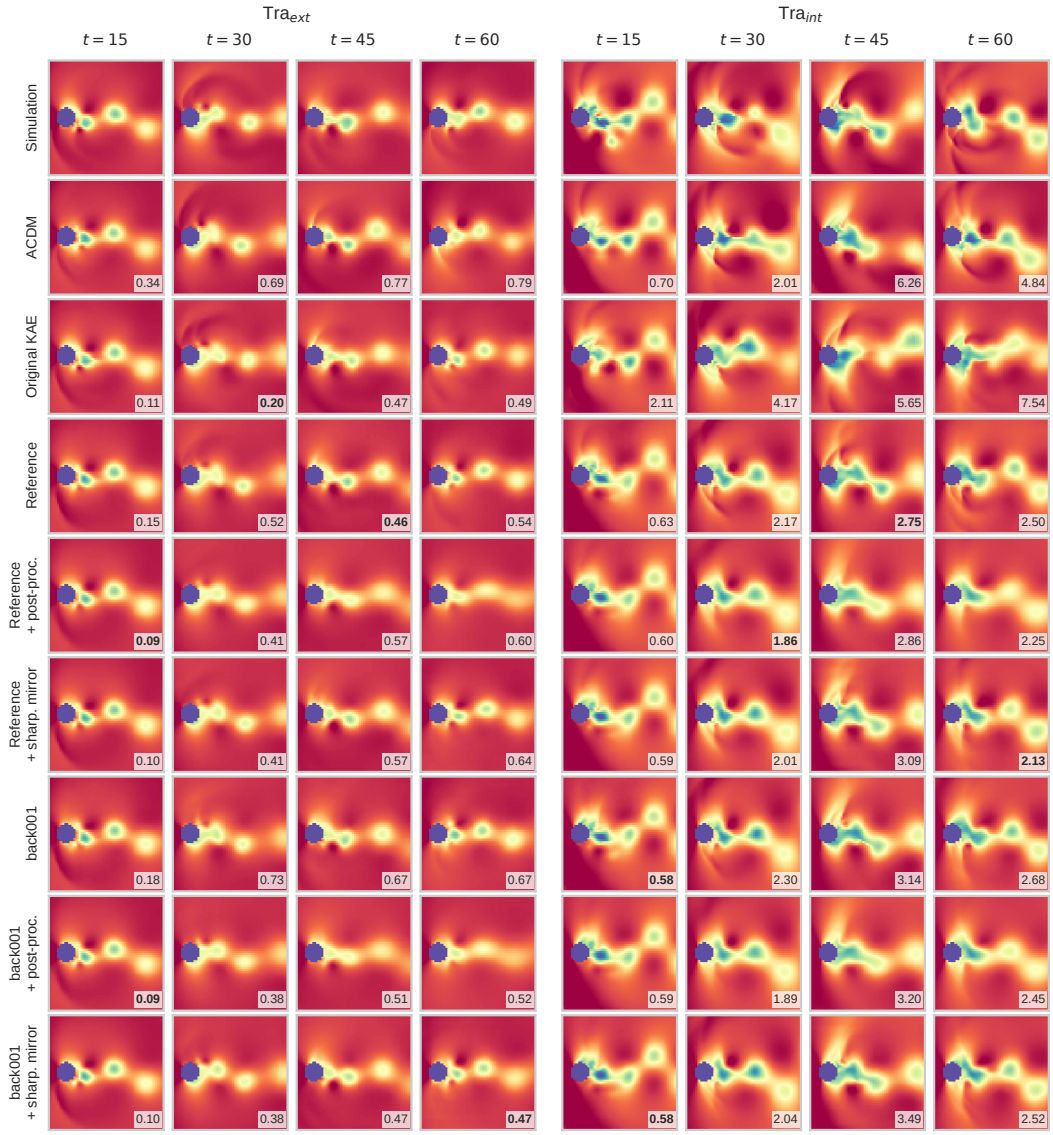


Figure 2: Pressure at $t = 15, 30, 45, 60$, first sequence of each test set (paper layout). CT-KAE rows without post-processing: seed 0; post-processing and sharpened mirror: 3-seed average. Number in each panel: pressure MSE ($\times 10^{-3}$) of that frame over the whole domain; bold is the best model in the column.

Table 9: Spectrum of $|v|$: relative error in the low ($k \leq 4$), mid (5–12) and high (13–32) wavenumber bands and the fitted mid-band slope.

	Tra_int (60 steps)				Tra_long (240 steps)			
	low	mid	high	slope	low	mid	high	slope
ACDM	0.03	0.02	0.04	-3.30	0.06	0.11	0.04	-3.27
U-Net$_{ut}$	0.03	0.02	0.02	-3.31	0.05	0.08	0.03	-3.38
FNO$_{16}$	0.09	0.14	0.27	-3.37	0.76	0.78	0.82	-3.56
Reference	0.20	0.09	0.11	-2.87	0.31	0.34	0.20	-2.53
Reference + post-proc.	0.20	0.19	0.25	-2.96	0.42	0.57	0.47	-2.36
back001	0.19	0.09	0.12	-2.96	0.28	0.41	0.24	-2.41
back001 + post-proc.	0.20	0.18	0.23	-3.00	0.41	0.58	0.46	-2.37
Ground truth				-3.30				-3.30

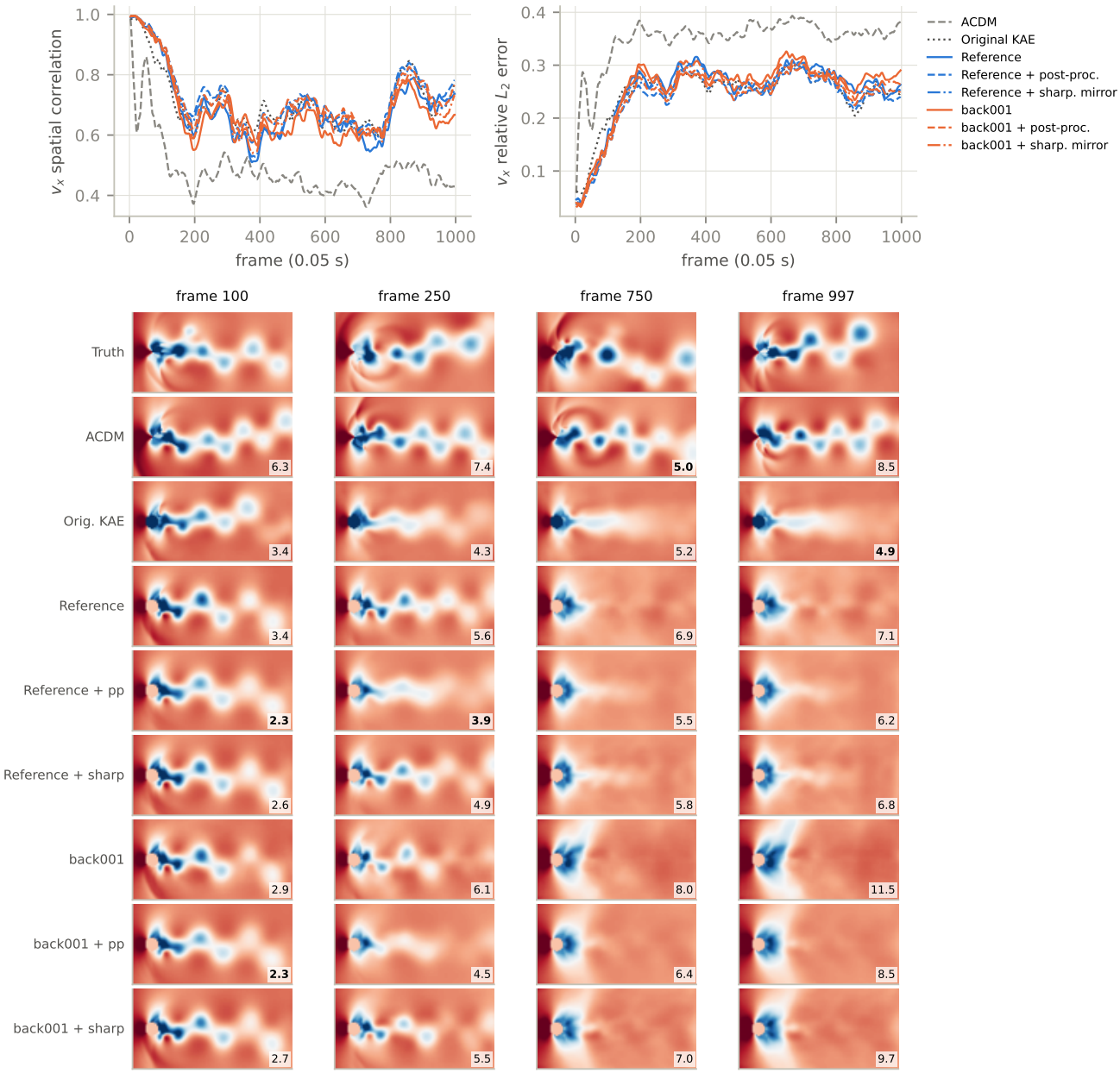


Figure 3: 1000-step rollout (Tra_long simulations, 0.05 s per frame, 2 trajectories). Top: v_x spatial correlation and relative L_2 error per frame. Bottom: pressure, first trajectory. Post-processing here is the 3-seed average of mixture + mirror, without shrinkage; "sharp" is the 3-seed sharpened mirror average. Number in each panel: pressure MSE ($\times 10^{-3}$) of that frame; bold is the best in the column.

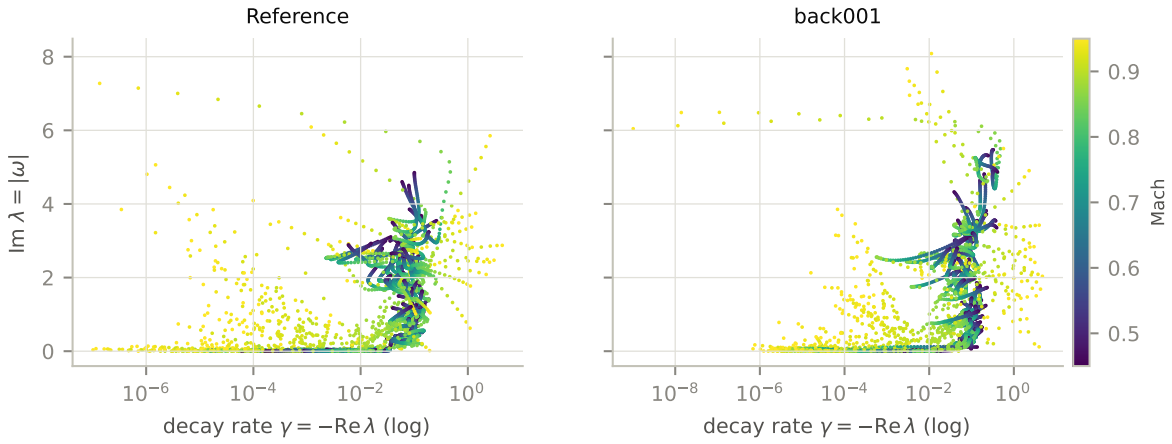


Figure 4: Generator eigenvalues $\lambda = -\gamma \pm i\omega$ for Mach 0.45–0.95 (seed 0).

Findings.

- **Integrators.** RK4 matches the exact exponential up to $\Delta t = 0.3$; explicit Euler diverges on long rollouts (from $\Delta t = 0.05$ for back001), as the oscillators are almost undamped.
- **Unseen time step.** Forecasts between the training time steps are as accurate as on them (Table 8).
- **Spectra.** The CT-KAE spectra are too shallow and its band errors are 3–7× those of ACDM and the U-Net (but well below FNO's on long rollouts); post-processing worsens them. The CT-KAE's MSE advantage on long rollouts comes from tracking phase, not from resolving small scales.
- **Not re-run:** the original KAE's loss and LoRA ablations need retraining; the CT-KAE's own ablation study (in the full report) replaces them.